Eigenvector alignment and eigenvalue shrinkage

$X_1, \dots X_n \in \mathbb{R}^d$ i.i.d, satisfying (i) $\mathbb{E} X_i = 0$, (ii) $\mathbb{E} X_i X_i^\top = \Sigma$, $\|\Sigma\|_{op} < C$,

(iii) $\mathbb{P}[|X_i^\top A X_i - \mathrm{Tr}\, A\Sigma| > t] \le C_p \left(\frac{\|A\|_F}{t}\right)^p$ $\forall$ deterministic $A \in \mathbb{R}^{d \times d}$.

Recall (Lecture 5): Let $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top \in \mathbb{R}^{d \times d}$.

If $n, d \to \infty$, $\frac{d}{n} \to \gamma \in (0, \infty)$, ESD of $\Sigma \to \nu$ weakly,

then ESD of $\hat{\Sigma} \to \mu_{MP}$ weakly a.s. Stieltjes transform

$m(z) = \frac{1}{\gamma} \underline{m}(z) + (1 - \frac{1}{\gamma})(-\frac{1}{z})$ of μ_{MP} satisfies

$$m(z) = \int \frac{1}{x(1 - \gamma - \gamma z m(z)) - z} \, d\nu(x) \iff z = -\frac{1}{\underline{m}(z)} + \gamma \int \frac{x}{1 + x\underline{m}(z)} \, d\nu(x)$$

For each $x \in \mathbb{R} \setminus \{0\}$, $\breve{m}(x) = \lim_{z \in \mathbb{C}^+ \to x} m(z)$ exists, and μ_{MP} has

density $f(x) = \frac{1}{\pi} \mathrm{Im}\, \breve{m}(x)$. [Silverstein, Choi '95]

Q: Can we improve over $\hat{\Sigma}$ as an estimator of Σ?

Let $\Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top$, $\hat{\Sigma} = \sum_{i=1}^d \hat{\lambda}_i \hat{v}_i \hat{v}_i^\top$.

Consider an eigenvalue shrinkage estimator, $\tilde{\Sigma} = \sum_{i=1}^d \ell_i \hat{v}_i \hat{v}_i^\top$

The Frobenius norm error is

$$\|\tilde{\Sigma} - \Sigma\|_F^2 = \|\tilde{\Sigma}\|_F^2 + \|\Sigma\|_F^2 - 2\,\mathrm{Tr}\,\tilde{\Sigma}\Sigma$$

$$= \sum_{i=1}^{d} \ell_i^2 + \|\Sigma\|_F^2 - 2\sum_{i=1}^{d} \ell_i \hat{v}_i^T \Sigma \hat{v}_i$$

$$= \sum_{i=1}^{d} \underbrace{\left(\ell_i^2 - 2\ell_i \hat{v}_i^T \Sigma \hat{v}_i \right)}_{\text{minimized at } \ell_i = \hat{v}_i^T \Sigma \hat{v}_i} + \|\Sigma\|_F^2$$

If Σ were known, "oracle" shrinkage estimator would set

$$\ell_i = \hat{v}_i^T \Sigma \hat{v}_i = \sum_{j=1}^{d} (\hat{v}_i^T v_j)^2 \lambda_j$$

Thm (Ledoit, Péché): Let $f: \mathbb{R} \to \mathbb{R}$ be continuous, bdd, compact support.

Let $\Delta_n(f) = \frac{1}{d}\sum_{i=1}^{d} \ell_i f(\hat{\lambda}_i) = \frac{1}{d}\sum_{i=1}^{d}\sum_{j=1}^{d} (\hat{v}_i^T v_j)^2 f(\hat{\lambda}_i)\lambda_j$

Under the assumptions of the M-P theorem, a.s. as $n, d \to \infty$,

$$\Delta_n(f) \to \Delta(f) = \int \underbrace{\frac{\lambda}{|1 - \gamma - \gamma\lambda \check{m}(\lambda)|^2}}_{:= \delta(\lambda)} f(\lambda)\, d\mu_{MP}(\lambda)$$

Interpretation: Consider $f(\lambda) \approx \mathbb{1}_{[a, a+\varepsilon]}(\lambda)$ for small ε. Then

$$\frac{1}{|\{i : \hat{\lambda}_i \in [a, a+\varepsilon]\}|} \sum_{i : \hat{\lambda}_i \in [a, a+\varepsilon]} \ell_i \approx \frac{\Delta_n(f)}{\frac{1}{d}|\{i : \hat{\lambda}_i \in [a, a+\varepsilon]\}|} \approx \frac{\Delta(f)}{\int_a^{a+\varepsilon} d\mu_{MP}(\lambda)} \approx \delta(a)$$

Thus: For $\hat{\lambda}_i \approx a$, the oracle value is $\ell_i \approx \delta(a)$.

This motivates the following eigenvalue shrinkage estimator of Σ:

(1) Estimate density $\hat{f}(x)$ of μ_{MP} from eigenvalues $\hat{\Sigma} = \sum_{i=1}^{d} \hat{\lambda}_i \hat{v}_i \hat{v}_i^T$

(2) Estimate $\breve{m}(\lambda_i)$ by $\hat{m}(\hat{\lambda}_i) = \int \frac{1}{x - z} \hat{f}(x) dx$ at $z = \hat{\lambda}_i + i\varepsilon$

(3) Estimate $S(\hat{\lambda}_i)$ by $\hat{S}(\hat{\lambda}_i) = \frac{\hat{\lambda}_i}{|1 - \gamma \hat{\lambda}_i \hat{m}(\hat{\lambda}_i)|^2}$

(4) Estimate Σ by $\tilde{\Sigma} = \sum_{i=1}^{d} \hat{S}(\hat{\lambda}_i) \hat{v}_i \hat{v}_i^T$.

(See Ledoit-Wolf '12 for more sophisticated procedure, which simultaneously estimates ESD of Σ and $\{\breve{m}(\hat{\lambda}_i)\}_{i=1}^{d}$.)

Proof idea of Thm: $\sum_{i,j=1}^{d} (\hat{v}_i^T v_j)^2 f(\hat{\lambda}_i) g(\lambda_j)$ may be analyzed via $\operatorname{Tr}(\hat{\Sigma} - zI)^{-1} g(\Sigma) = \sum_{i,j=1}^{d} (\hat{v}_i^T v_j)^2 \frac{1}{\hat{\lambda}_i - z} g(\lambda_j)$.

Lemma: Let $A \in \mathbb{C}^{d \times d}$ be deterministic, $\|A\|_{op} < C$. For any $z \in \mathbb{C}^+$,

$$\frac{1}{d} \operatorname{Tr}(\hat{\Sigma} - zI)^{-1} A - \frac{1}{d} \operatorname{Tr}(-zI - z\underline{m}(z)\Sigma)^{-1} A \to 0 \text{ a.s.}$$

[$(-zI - z\underline{m}(z)\Sigma)^{-1}$ is a <u>deterministic equivalent</u> for $R(z) = (\hat{\Sigma} - zI)^{-1}$.]

Proof: Let $z = \lambda + i\eta \in \mathbb{C}^+$, $R = (\hat{\Sigma} - zI)^{-1}$, $R^{(j)} = (\frac{1}{n} \sum_{i \neq j} x_i x_i^T - zI)^{-1}$

Recall (Lecture 5): For any $M \in \mathbb{C}^{d \times d}$:

$$\frac{1}{d} \operatorname{Tr} M = \frac{1}{d} \operatorname{Tr} R(\hat{\Sigma} - zI) M$$

$$= -\frac{z}{d}\operatorname{Tr} RM + \frac{1}{d}\sum_{i=1}^{n} \frac{\frac{1}{n} x_i^T M R^{(i)} x_i}{1+\frac{1}{n} x_i^T R^{(i)} x_i}$$

Take $M = I \Rightarrow -z \cdot m_n(z) = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{1+\frac{1}{n} x_i^T R^{(i)} x_i}$ (Lecture 5).

Apply (Lecture 5):

- $m_n(z) \to m(z)$ a.s.
- $\sup_{i=1}^{n} \left| \frac{1}{n} x_i^T R^{(i)} x_i - \frac{1}{n}\operatorname{Tr} R^{(i)}\Sigma \right| \le n^{-1+\varepsilon} \|R^{(i)}\|_F \le n^{-1/2+\varepsilon} \cdot \frac{1}{\eta}$ a.s. for all large n, d
- $\left| \frac{1}{n}\operatorname{Tr} R^{(i)}\Sigma - \frac{1}{n}\operatorname{Tr} R\Sigma \right| \le \frac{C}{n\eta}$
- $\left| 1+\frac{1}{n} x_i^T R^{(i)} x_i \right| \ge \frac{\eta}{|z|}$

$$\Rightarrow -z \cdot m(z) = \frac{1}{1+\frac{1}{n}\operatorname{Tr} R\Sigma} + r_n, \quad r_n \to 0 \text{ a.s.}$$

$$\Rightarrow \frac{1}{d}\operatorname{Tr} R\Sigma = \frac{n}{d}\left(\frac{1}{-z m(z) - r_n} - 1\right)$$

$$\to -\frac{1}{\gamma}\left(\frac{1}{z m(z)} + 1\right) \text{ a.s.,} \qquad (**)$$

$$\sup_{i=1}^{n} \left| 1+\frac{1}{n} x_i^T R^{(i)} x_i - \left(-\frac{1}{z m(z)}\right) \right| \to 0 \text{ a.s.}$$

Then for any deterministic $M \in \mathbb{C}^{d \times d}$ w/ $\|M\|_{op} < C$:

- $\sup_{i=1}^{n} \left| \frac{1}{n} x_i^T M R^{(i)} x_i - \frac{1}{n}\operatorname{Tr} M R^{(i)}\Sigma \right| \le n^{-1/2+\varepsilon} \cdot \frac{C}{\eta}$ a.s. large n, d
- $\left| \frac{1}{n}\operatorname{Tr} M R^{(i)}\Sigma - \frac{1}{n}\operatorname{Tr} M R\Sigma \right| \le \frac{C}{n\eta}$.

$$\Rightarrow \frac{1}{p}\operatorname{Tr} M = -\frac{z}{p}\operatorname{Tr} RM + \frac{n}{p}\left(\frac{1}{n}\operatorname{Tr} M\Sigma\right)\left(-z\underline{m}(z)\right) + v_n$$

$$= \frac{1}{p}\operatorname{Tr} R\left(-zI - z\underline{m}(z)\Sigma\right)M + v_n, \quad v_n \to 0 \text{ a.s.}$$

Recall (Lecture 5) $|z(1+\underline{m}_n(z)\cdot t)| \geq \eta \quad \forall t \geq 0$

$$\Rightarrow |z(1+\underline{m}(z)\cdot t)| \geq \eta \quad \forall t \geq 0 \Rightarrow \|(zI + z\underline{m}(z)\Sigma)^{-1}\|_{op} \leq \frac{1}{\eta}.$$

Take $M = (-zI - z\underline{m}(z)\Sigma)^{-1}A$ for the given $A \in \mathbb{C}^{p\times n}$. ∎

Cor: Let $g: \mathbb{R}\to\mathbb{R}$ be bdd, $g(\Sigma) = \sum_{j=1}^{p} g(\lambda_j) v_j v_j^T$. Then

$$\frac{1}{p}\operatorname{Tr}(zI - \Sigma)^{-1} g(\Sigma)$$

$$\to \int \frac{g(x)}{-z - z\underline{m}(z)x}\, dH(x) = \int \frac{g(x)}{x(1-\gamma-\gamma z\underline{m}(z)) - z}\, dH(x) \quad \text{a.s.}$$

[For $g(x)=1$, $g(\Sigma)=I$, this recovers the M-P Theorem.

For $g(x)=x$, $z = -\frac{1}{\underline{m}(z)} + \gamma\int \frac{x}{1+x\underline{m}(z)}\, dH(x)$

$$\Rightarrow \int \frac{g(x)}{-z - z\underline{m}(z)x}\, dH(x) = -\frac{1}{z}\int \frac{x}{1+x\underline{m}(z)}\, dH(x)$$

$$= -\frac{1}{z}\left(\frac{1}{z\underline{m}(z)} + 1\right), \text{ recovering (**).} \quad]$$

Proof: Apply lemma w/ $A = g(\Sigma)$, and observe

$$\frac{1}{p}\operatorname{Tr}(-zI - z\underline{m}(z)\Sigma)^{-1} g(\Sigma) \to \int \frac{g(x)}{-z - z\underline{m}(z)x}\, dH(x).$$

Now $\underline{m}(z) = \gamma m(z) + (1-\gamma)(-\frac{1}{z}) \Rightarrow -z\underline{m}(z) = 1 - \gamma - \gamma z m(z)$. ∎

Proof of Thm: $\Delta_n(f) = \frac{1}{d}\sum_{i,j=1}^{d} (\hat{v}_i^\top v_j)^2 f(\hat{\lambda}_i)\lambda_j$, $f \in C_0(\mathbb{R})$.

Recall (Lecture 3):

$$\sup_{\hat{\lambda}\in\mathbb{R}} \left| f(\hat{\lambda}) - \int f(x)\cdot \frac{1}{\pi}\frac{\eta}{(\hat{\lambda}-x)^2+\eta^2}\,dx \right| \le \varepsilon(\eta) \to 0 \text{ as } \eta \to 0.$$

Let $\Delta_n^\eta(f) = \frac{1}{d}\sum_{i,j=1}^{d} (\hat{v}_i^\top v_j)^2 \int f(x)\cdot \frac{1}{\pi}\frac{\eta}{(\hat{\lambda}_i-x)^2+\eta^2}\,dx \cdot \lambda_j$.

$$\Rightarrow \left|\Delta_n^\eta(f) - \Delta_n(f)\right| \le \varepsilon(\eta)\cdot \underbrace{\frac{1}{d}\sum_{i,j=1}^{d} (\hat{v}_i^\top v_j)^2 \lambda_j}_{= \frac{1}{d}\sum_{i=1}^{d} \hat{v}_i^\top \Sigma \hat{v}_i = \frac{1}{d}\mathrm{Tr}\,\Sigma} \le C\cdot\varepsilon(\eta)$$

Note: $\Delta_n^\eta(f) = \frac{1}{d}\sum_{i,j=1}^{d} (\hat{v}_i^\top v_j)^2 \int f(x)\cdot \frac{1}{\pi}\mathrm{Im}\,\frac{1}{\hat{\lambda}_i - (x+i\eta)}\,dx\cdot\lambda_j$

$$= \int f(x)\cdot\frac{1}{\pi}\mathrm{Im}\,\underbrace{\frac{1}{d}\mathrm{Tr}\left(\hat{\Sigma}-(x+i\eta)I\right)^{-1}\Sigma}_{\to -\frac{1}{\gamma}\left(\frac{1}{z\bar{m}(z)}+1\right),\ z = x+i\eta}\,dx$$

$$\to \int f(x)\cdot\frac{1}{\pi}\mathrm{Im}\left[-\frac{1}{\gamma}\left(\frac{1}{z\bar{m}(z)}+1\right)\right]dx \quad \text{as } n,d\to\infty$$

$$= \int f(x)\cdot\frac{1}{\pi}\mathrm{Im}\left[\frac{1}{\gamma}\cdot\frac{1}{1-\gamma-\gamma z m(z)}\right]dx$$

$$= \underbrace{\int f(x)\cdot\frac{1}{\pi}\frac{\mathrm{Im}\,z m(z)}{|1-\gamma-\gamma z m(z)|^2}\,dx}_{:=\Delta^\eta(f)}$$

As $\eta\to 0$: $\Delta^\eta(f) \to \int f(x)\frac{x}{|1-\gamma-\gamma x\breve{m}(x)|^2}\cdot\frac{1}{\pi}\mathrm{Im}\,\breve{m}(x)\,dx$

$$= \int f(x) \cdot \underbrace{\frac{x}{|1-\gamma-\gamma x \check{m}(x)|^2}}_{\delta(x)} d\mu_{MP}(x) = \Delta(f).$$

$$\Rightarrow \limsup_{n,d\to\infty} |\Delta_n(f) - \Delta(f)|$$

$$\le \limsup_{n,d\to\infty} |\Delta_n(f) - \Delta_n^\eta(f)| + |\Delta_n^\eta(f) - \Delta^\eta(f)| + |\Delta^\eta(f) - \Delta(f)|$$

$$\le \varepsilon(\eta) \text{ a.s., where } \varepsilon(\eta) \to 0 \text{ as } \eta \to 0.$$

$$\Rightarrow \Delta_n(f) \to \Delta(f) \text{ a.s.}$$

Generalization error of ridge regression

Let $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^d \times \mathbb{R}$ be i.i.d.,

$$\hat{\beta} = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \|\beta\|_2^2 \qquad (\lambda > 0)$$

Given $x_{test} \in \mathbb{R}^d$, the ridge regression predictor of y_{test} is $x_{test}^T \hat{\beta}$.

Q: How large is the generalization error $\mathbb{E}(y_{test} - x_{test}^T \hat{\beta})^2$, and how does this depend on the distribution of (x_i, y_i)?

Explicitly: Let $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$, $y = (y_1, \dots, y_n) \in \mathbb{R}^n$.

$$\hat{\beta} = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \frac{1}{n} \|y - X^T \beta\|_2^2 + \lambda \|\beta\|_2^2$$

$$= \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \beta^T \left(\frac{1}{n} XX^T + \lambda I\right) \beta + \frac{2}{n} \beta^T X y + \frac{1}{n} \|y\|_2^2$$

$$= \underbrace{\left(\frac{1}{n} XX^T + \lambda I\right)^{-1}}_{= R(-\lambda), \text{ resolvent of } \hat{\Sigma} = \frac{1}{n} XX^T \text{ at } z = -\lambda.} \cdot \frac{1}{n} X y$$

Suppose: (i) X_i satisfies conditions of M-P Thm, $\Sigma = \sum_{i=1}^{d} \lambda_i v_i v_i^T$

($\mathbb{E} X_i = 0$, $\mathbb{E} X_i X_i^T = \Sigma$, $\frac{d}{n} \to \gamma$, ESD of $\Sigma \to \nu$ weakly)

(ii) $y_i = \beta_*^T x_i + \varepsilon_i$, ε_i independent of x_i w/ $\mathbb{E}\varepsilon_i = 0$, $\mathbb{E}\varepsilon_i^2 = \sigma^2$

(iii) $\beta_* \in \mathbb{R}^d$ random w/ $\mathbb{E}\beta_* \beta_*^T = \frac{1}{d}\Sigma_\beta$, $\|\Sigma_\beta\|_{op} < C$,

$\mathbb{E}\|\beta_*\|_2^2 = \frac{1}{d}\mathrm{Tr}\,\Sigma_\beta \to r^2$, $\sum_{i=1}^{d} \frac{v_i^T \Sigma_\beta v_i}{\mathrm{Tr}\,\Sigma_\beta} \cdot \delta_{\lambda_i} \to \nu_\beta$ weakly.

ν_β represents "alignment" of β_* w/ eigenvectors of Σ.

Special cases: $\Sigma = I$ (isotropic data). Then $\nu = \nu_\beta = \delta_1$.

$\Sigma_\beta = r^2 \cdot I$ (signal unaligned w/ Σ). Then $\nu_\beta = \nu$.

Thm (Wu, Xu '20, Hastie et al '22): Almost surely as $n, d \to \infty$,

$$\mathbb{E}\left(y_{test} - x_{test}^T \hat{\beta}\right)^2 \to \mathcal{B} + \mathcal{V} + \sigma^2 \text{ where}$$

$$\mathcal{B} = r^2 \cdot \frac{\underline{m}'(-\lambda)}{\underline{m}(-\lambda)^2} \int \frac{x}{(1 + x\,\underline{m}(-\lambda))^2}\, d\nu_\beta(x)$$

$$\mathcal{V} = \sigma^2 \left(\frac{\underline{m}'(-\lambda)}{\underline{m}(-\lambda)^2} - 1 \right)$$

Proof: Write $\hat{\beta} = \underbrace{\left(\frac{1}{n}XX^T + \lambda I\right)^{-1} \cdot \frac{1}{n} X X^T \beta_*}_{:= \bar{\beta}} + \underbrace{\left(\frac{1}{n}XX^T + \lambda I\right)^{-1} \cdot \frac{1}{n} X \varepsilon}_{:= \bar{\varepsilon}}$

Then $\mathbb{E}\left(y_{test} - x_{test}^T \hat{\beta}\right)^2$

$= \mathbb{E}\left(x_{test}^T (\beta_* - \hat{\beta}) + \varepsilon_{test}\right)^2$

$$= \mathbb{E}\left[(\hat{\beta} - \beta_*)^T \Sigma (\hat{\beta} - \beta_*)\right] + \sigma^2$$

$$= \underbrace{\mathbb{E}\left[(\bar{\beta} - \beta_*)^T \Sigma (\bar{\beta} - \beta_*)\right]}_{\text{squared bias}} + \underbrace{\mathbb{E}\left[\bar{\varepsilon}^T \Sigma \bar{\varepsilon}\right]}_{\text{variance}} + \underbrace{\sigma^2}_{\text{intrinsic error}}$$

Let $\hat{\Sigma} = \frac{1}{n} X X^T$, $R = (\hat{\Sigma} + \lambda I)^{-1}$. Then $I = R(\hat{\Sigma} + \lambda I)$, and

$$\text{squared bias} = \mathbb{E}\left[\beta_*^T (\hat{\Sigma} R - I) \Sigma (R\hat{\Sigma} - I) \beta_*\right]$$

$$= \mathbb{E}\left[\lambda^2 \cdot \beta_*^T R \Sigma R \beta_*\right] = \frac{\lambda^2}{d} \mathbb{E}\, \mathrm{Tr}\, R \Sigma R \Sigma_\beta.$$

$$\text{variance} = \mathbb{E}\left[\frac{1}{n^2} \varepsilon^T X^T R \Sigma R X \varepsilon\right]$$

$$= \mathbb{E}\left[\frac{\sigma^2}{n^2} \mathrm{Tr}\, X^T R \Sigma R X\right] = \frac{\sigma^2}{n} \mathbb{E}\, \mathrm{Tr}\, R \Sigma R \hat{\Sigma}$$

① Variance: $\mathrm{Tr}\, R \Sigma R \hat{\Sigma} = \mathrm{Tr}\, R \Sigma (I - \lambda R)$

$$= \mathrm{Tr}\, R\Sigma - \lambda\, \mathrm{Tr}\, R^2 \Sigma.$$

For $z \in \mathbb{C}^+$: $\frac{1}{d} \mathrm{Tr}\, R(z) \Sigma \to -\frac{1}{\gamma}\left(\frac{1}{z\underline{m}(z)} + 1\right)$ a.s.

Both sides are analytic, locally bdd. over $z \in \mathbb{C} \setminus [0, \infty)$. Thus:

$$\frac{1}{d} \mathrm{Tr}\, R(z) \Sigma \to -\frac{1}{\gamma}\left(\frac{1}{z\underline{m}(z)} + 1\right) \quad \forall z \in \mathbb{C} \setminus [0, \infty)$$

$$\underbrace{\frac{d}{dz}\left[\frac{1}{d} \mathrm{Tr}\, R(z) \Sigma\right]}_{= \frac{1}{d} \mathrm{Tr}\, R(z)^2 \Sigma,} \to \underbrace{\frac{d}{dz}\left[-\frac{1}{\gamma}\left(\frac{1}{z\underline{m}(z)} + 1\right)\right]}_{= \frac{\underline{m}(z) + z\underline{m}'(z)}{\gamma z^2 \underline{m}(z)^2}}$$

b/c $\frac{d}{dz} R(z) = \frac{d}{dz} (\hat{\Sigma} - zI)^{-1} = R(z)^2$

So for $z = -\lambda$, by dominated convergence:

$$\text{variance} = \frac{\sigma^2 d}{n} \cdot \frac{1}{d} \mathbb{E}\, \mathrm{Tr}\, R\Sigma R\hat{\Sigma}$$

$$\to -\sigma^2\left(\frac{1}{-\lambda \underline{m}(-\lambda)} + 1\right) - \sigma^2\lambda\left(\frac{\underline{m}(-\lambda) - \lambda \underline{m}'(-\lambda)}{\lambda^2 \underline{m}(-\lambda)^2}\right)$$

$$= \sigma^2\left(\frac{\underline{m}'(-\lambda)}{\underline{m}(-\lambda)^2} - 1\right) =: V.$$

② Bias: For $z \in \mathbb{C}^+$, write

$$\frac{1}{d}\mathrm{Tr}\, R\Sigma_\beta = \frac{1}{d}\mathrm{Tr}\, R\Sigma_\beta R(\hat{\Sigma} - zI)$$

$$= -\frac{z}{d}\mathrm{Tr}\, R^2\Sigma_\beta + \frac{1}{d}\sum_{i=1}^n \frac{1}{n} x_i^T R\Sigma_\beta R x_i$$

$$= -\frac{z}{d}\mathrm{Tr}\, R^2\Sigma_\beta + \frac{1}{d}\sum_{i=1}^n \frac{\frac{1}{n} x_i^T R^{(i)}\Sigma_\beta R^{(i)} x_i}{(1 + \frac{1}{n} x_i^T R^{(i)} x_i)^2}$$

(by Sherman-Morrison, Lecture 5)

Apply

- $\sup_{n \ge 1} \left| \frac{1}{n} x_i^T R^{(i)}\Sigma_\beta R^{(i)} x_i - \frac{1}{n}\mathrm{Tr}\, R\Sigma_\beta R\Sigma \right| \to 0$ a.s.
- $\sup_{n \ge 1} \left| 1 + \frac{1}{n} x_i^T R^{(i)} x_i - \left(-\frac{1}{z\underline{m}(z)}\right) \right| \to 0$ a.s.,

  and $\left| 1 + \frac{1}{n} x_i^T R^{(i)} x_i \right| \ge \frac{\eta}{|z|}$.

$$\Rightarrow \frac{1}{d}\sum_{i=1}^n \frac{\frac{1}{n} x_i^T R^{(i)}\Sigma_\beta R^{(i)} x_i}{(1 + \frac{1}{n} x_i^T R^{(i)} x_i)^2} - (z\underline{m}(z))^2 \cdot \frac{1}{d}\mathrm{Tr}\, R\Sigma_\beta R\Sigma \to 0 \text{ a.s.}$$

$$\Rightarrow \frac{1}{d}\operatorname{tr} R\Sigma_\beta R\Sigma - \left(\frac{1}{z\underline{m}(z)}\right)^2\left(\frac{1}{d}\operatorname{tr} R\Sigma_\beta + \frac{p}{n}\operatorname{tr} R^2\Sigma_\beta\right)$$

Here:

- $\frac{1}{d}\operatorname{tr} R\Sigma_\beta - \frac{1}{d}\operatorname{tr}(-zI - z\underline{m}(z)\Sigma)^{-1}\Sigma_\beta \to 0$ a.s., where

$$\frac{1}{d}\operatorname{tr}(-zI - z\underline{m}(z)\Sigma)^{-1}\Sigma_\beta = \frac{1}{d}\operatorname{tr}\sum_{i=1}^{p}\frac{1}{-z - z\underline{m}(z)\lambda_i}v_i v_i^T\Sigma_\beta$$

$$= \sum_{i=1}^{p}\frac{v_i^T\Sigma_\beta v_i}{-z - z\underline{m}(z)\lambda_i}\cdot\frac{1}{p}\operatorname{tr}\Sigma_\beta$$

$$\to \int \frac{1}{-z - z\underline{m}(z)x}\,d\nu_\beta(x)\cdot r^2$$

- $\frac{1}{d}\operatorname{tr} R\Sigma_\beta + \frac{p}{n}\operatorname{tr} R^2\Sigma_\beta = \frac{d}{p}\left[\frac{zp}{n}\operatorname{tr} R\Sigma_\beta\right]$

$$\to \frac{d}{p}\left[\int -\frac{1}{1+x\underline{m}(z)}\,d\nu_\beta(x)\cdot r^2\right]$$

$$= r^2\cdot\int\frac{x\underline{m}'(z)}{(1+x\underline{m}(z))^2}\,d\nu_\beta(x)$$

For $z = -\lambda$, by dominated convergence,

$$\text{Squared-bias} = \frac{\lambda^2}{d}\mathbb{E}\operatorname{tr} R\Sigma_\beta R\Sigma$$

$$\to \frac{r^2\underline{m}'(-\lambda)}{\underline{m}(-\lambda)^2}\int\frac{x}{(1+x\underline{m}(-\lambda))^2}\,d\nu_\beta(x) =: \mathcal{B}.$$

∎

Problem 6: Suppose $\Sigma = I$. Let $m(z) = \frac{1}{\gamma} \underline{m}(z) + (1 - \frac{1}{\gamma})(-\frac{1}{z})$.

(a) For any $\lambda > 0$, show that equivalently

$$\mathcal{B} = r^2 \cdot \lambda^2 m'(-\lambda)$$

$$\mathcal{V} = \sigma^2 \gamma [m(-\lambda) - \lambda m'(-\lambda)]$$

(b) Suppose $\gamma \neq 1$, and consider the ridgeless limit $\lambda \to 0$.

Derive simple expressions for $\lim_{\lambda \to 0} \mathcal{B}$ and $\lim_{\lambda \to 0} \mathcal{V}$,

in both cases $\gamma < 1$ and $\gamma > 1$. Plot these limits as

a function of γ.

(c) Let $M^\dagger$ be the Moore-Penrose pseudo-inverse, and consider

$$\hat{\beta}_{\text{ridgeless}} = (XX^T)^\dagger X y = \begin{cases} \underset{\beta}{\operatorname{argmin}} \|y - X^T \beta\|_2^2 & \text{if } n > d \\ \underset{\beta : y = X^T \beta}{\operatorname{argmin}} \|\beta\|_2^2 & \text{if } n < d \end{cases}$$

Simulate the generalization error $\mathbb{E}(x_{\text{test}}^T \hat{\beta}_{\text{ridgeless}} - y_{\text{test}})^2$

and plot this as a function of γ.

How closely is this predicted by $\lim_{\lambda \to 0} \mathcal{B} + \mathcal{V} + \sigma^2$?